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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHE 521 — Past Questions & Materials
by Renance

Course: CHE 521
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 500

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Paper 1 of 2 — CHE 521 - Loss prevention in process industries

Level 500 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EYURON

COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF CHEMICAL ENGINEERING

Second Semester Examination 2024/2025 Session

COURSE TITLE: LOSS PREVENTION IN PROCESS INDUSTRIES

COURSE CODE: CHE 5321

Credit Unit: 3 Time Allowed: 3 HOURS

INSTRUCTION: Answer Five (5) questions only. All questions carry equal marks.

Question 1. (14 marks)

- a. Discuss five major types of hazards encountered in chemical process industries, For each type, give at least one specific example and explain how it can be controlled or mitigated, (8 marks)
- b. List four common sources of hazards in a process plant, (2 marks)
- c. Explain how human factors contribute to accidents and state two preventive measures. (3 marks)
- d. Itemize any two international safety standards relevant to process industries, (1 marks)

Question 2. (14 marks)

- a. Summarize the key lessons learned from the Bhopal disaster and explain how a strong safety culture and emergency preparedness could have reduced the impact. (3 marks)
- b. Explain the importance of plant maintenance in minimizing operational losses. Highlight three key benefits of a well-implemented maintenance program. (5 marks)
- c. Examine the difference and similarity between predictive, preventive, and corrective maintenance. Provide one example of each. (6 marks)

Question 3. (14 marks)

- a. An industrial plant suffered a major shutdown due to unexpected compressor failure.
 - i. Identify which type of maintenance could have prevented this loss and why, (2 marks)
 - ii. Recommend two steps to improve future maintenance planning. (2 marks)
- b. Describe the major steps involved in proper industrial waste management, (5 marks)
- c. Explain the legal implications of environmental non-compliance in process industries. Mention two key regulatory agencies or laws, (5 marks)

Question 4. (14 marks)

- a. A maintenance technician receives an electrical shock while working on a pump motor.
 - i. Identify three likely sources of electrical hazards in a chemical plant. (3 marks)
 - ii. Recommend four specific prevention measures based on recognized standards. (2 marks)
 - iii. Cite two relevant electrical safety regulations or standards. (2 marks)
- b. Chemical and environmental hazards present significant risks in process industries. ,
 - i. Describe four types of chemical hazards and their common routes of exposure. (4 marks)
 - ii. Outline three key controls to prevent environmental hazards such as spills, leaks, and emission.

(3 marks)

Question 5: (14 marks)

- Define what constitutes a hazardous area in industrial settings and explain why proper classification is essential for industrial safety. (4 marks)
- Discuss the similarity and difference among the Zone Classification (IEC/ATEX) and Division Classification (NEC/NFPA) systems for hazardous areas. (7marks)
- Give examples of when each zone/division would apply. (3 marks)

Question 6. (14 marks)

A petrochemical plant experienced a serious accident where a worker suffered chemical burns and inhalation injuries during a reactor cleaning operation. The first aid response was delayed due to inadequate training of nearby personnel, and the first aid kit lacked appropriate supplies for chemical

exposures. — . ae

- Analyze the failures in the first aid system that contributed to the severity of this incident. (6 marks)
 - Recommend specific improvements to prevent similar incidents, aid response and long-term system enhancements. (4 marks)
 - Explain how this incident. could have been prevented through better integration 0 planning with operational risk assessment. (4 marks)
- addressing both immediate first

f first aid

Question 7: (14 marks)

- Define Passive Fire Protection (PFP) and Active Fire Protection (AFP), explaining the . fundamental differences between these two approaches to fire safety. (2 marks).
- Compare and contrast the following fire protection elements by categorizing them as either passive or active, and explain their specific functions:

- Fire-rated walls
- Sprinkler systems
- Fire doors
- Smoke detectors
- Intumescent coatings
- Fire extinguishers (8 marks) —

- A data center requires comprehensive fire protecti active systems, justifying your choices based on the spec facility. (4 marks)

- on. Recommend a combination of passive and ific hazards and requirements of this

Paper 2 of 2 — CHE 521 SAFETY LECTURE NOTE

Level 500 • Second Semester • Handout

1. Safe Work Practices

Safe work practices are essential in any workplace, especially in industries that involve hazardous materials and equipment. It is important to be aware of the potential hazards, and take measures to prevent accidents and injuries. Safe work practices are written methods that outline how to perform tasks with minimal risk to people, equipment, and the environment. They are proactive strategies to prevent injuries, illnesses, and incidents at the workplace.

Fundamentals in safe work practices:

- Hazard Identification and Risk Assessment (HIRA)
- Standard Operating Procedures (SOPs)
- Job Safety Analysis (JSA)
- Emergency Preparedness

Hazard Identification and Risk Assessment (HIRA):

HIRA is a systematic process used to identify potential hazards in the workplace and assess the risks associated with them. It helps in determining the likelihood and severity of incidents, enabling effective control measures to be implemented to minimize risks and ensure worker safety.

Standard Operating Procedures (SOPs):

SOPs are detailed, written instructions designed to ensure consistency and quality in performing specific tasks. They help maintain safety, efficiency, and compliance with regulations by providing clear guidelines for workers to follow.

Job Safety Analysis (JSA):

JSA is a method that breaks down a job into individual steps to identify hazards and implement controls at each stage. It enhances workplace safety by proactively addressing potential risks before tasks are performed.

Emergency Preparedness:

Emergency preparedness involves planning and implementing measures to respond effectively to emergencies such as fires, chemical spills, or natural disasters. It includes training, emergency drills, and having proper equipment and procedures in place to protect lives and property.

Examples of safe work practices:

- Lockout/Tagout (LOTO)
- Permit-to-Work (PTW)
- Toolbox Talks

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- Buddy System

Benefits of Safe work practices:

- ☐ Reduces workplace incidents
- ☐ Encourages a culture of safety
- ☐ Ensures regulatory compliance (e.g., OSHA, ISO 45001)

2. Personal Protective Equipment (PPE) and Costumes

PPE includes all wearable items used to protect individuals from exposure to occupational hazards. In chemical engineering, PPE is the last line of defense when engineering and administrative controls are insufficient.

Categories and Examples:

- ☐ Head: Hard hats
- ☐ Eye & Face: Safety goggles, face shields
- ☐ Respiratory: Dust masks, respirators
- ☐ Hearing: Ear plugs, ear muffs
- ☐ Hand: Rubber, nitrile, or heat-resistant gloves
- ☐ Body: Aprons, chemical-resistant overalls
- ☐ Foot: Steel-toed or chemical-resistant boots

Proper Use and Limitations: Choose the correct PPE and ensure proper fit and maintenance.

PPE is a supplement, not a substitute, for engineering controls

3. Electrical Safety

Common Hazards: Shock and Electrocution, Arc Flash/Blast, and Fire and Explosions

Importance: Electrical safety is crucial to prevent electrical shock and electrocution.

Sources of Common electrical Hazards in Chemical Plants:

- ☐ Motors and pumps
- ☐ Control panels
- ☐ Heaters and ovens

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Prevention Measures:

- ☐ Proper grounding, insulation
- ☐ Use insulated tools and electrical-rated gloves
- ☐ Follow lockout/tagout (LOTO) procedures.
- ☐ Maintain safe clearances
- ☐ Conduct regular audits and maintenance

Regulations:

- OSHA 1910 Subpart S
- NFPA 70E

4. Mechanical Hazards

Mechanical hazards arise from moving machinery parts or stored mechanical

energy that may cause injury.

Common Sources:

- Rotating equipment
- Pinch points
- Sharp surfaces
- Pressure vessels

Types of Mechanical Hazards: Crushing (Crushed limbs), cutting, entanglement hazards, abrasions, Broken bones and amputations

Controls / Prevention Measures: Machine Guarding, proper maintenance, Emergency stops, Operator training, and safe operating procedures.

5. Chemical Hazards

Chemical hazards refer to hazards caused by chemicals and substances.

Types of Chemical Hazards: Toxicity, flammability, and reactivity hazards, Corrosivity, and reactivity.

Routes of Exposure: Inhalation, Ingestion, Dermal absorption and Injection

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Controls / Prevention Measures: Proper handling, storage, and disposal of chemicals.

Engineering controls (ventilation), Administrative controls (training), PPE (gloves, goggles), Labeling and MSDS and Spill response protocols

6. Environmental Hazards

Definition: Environmental hazards refer to hazards that affect the environment and human health. They are hazards to the environment from emissions, waste, or leaks.

Types of Environmental Hazards: Air and water pollution, climate change, and waste management.

Sources Environmental Hazards: Volatile organic compound (VOCs) emissions, Spills and leaks, Waste mismanagement and Noise and heat discharge

Impact are: Air and water pollution, Soil degradation, and Biodiversity loss

Controls / Prevention Measures: Sustainable practices, waste reduction, and pollution control, Treatment plants, Monitoring systems, Emergency containment, and Regulatory compliance (NESREA, EPA)

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Examination Questions and Answers

Short Answer Questions

1. What is the importance of PPE in the workplace? (5 marks)
2. Describe the types of electrical hazards and prevention measures. (10 marks)
3. What are the types of mechanical hazards and how can they be prevented? (10

marks)

Long Answer Questions

1. Discuss the importance of safe work practices in the chemical industry. (20 marks)
2. Describe the types of chemical hazards and prevention measures. (20 marks)
3. Discuss the importance of environmental hazard prevention and control. (20 marks)

Marking Scheme

- Short answer questions: 5-10 marks each, depending on complexity.
- Long answer questions: 20 marks each, depending on complexity.
- Marks awarded for: Accuracy and completeness of knowledge, clarity and organization of writing and use of relevant examples and diagrams.

Example Answers

Short Answer Question 1

What is the importance of PPE in the workplace? (5 marks)

Answer

5

PPE is essential to prevent injuries and illnesses caused by hazardous materials and equipment. It provides a barrier between the worker and the hazard, reducing the risk of exposure.

Long Answer Question 1

Discuss the importance of safe work practices in the chemical industry. (20 marks)

Answer

Safe work practices are crucial in the chemical industry to prevent accidents and injuries. Chemical engineers must be aware of the potential hazards and take measures to prevent them. This includes proper handling and storage of chemicals, use of PPE, and following safe operating procedures. By implementing safe work practices, chemical engineers can minimize risks and ensure a safe working environment.

6

Safety in Hazardous Areas: Industrial Safety and Analysis

Introduction

Safety in hazardous areas is a critical aspect of industrial safety, particularly in the chemical process industry. Hazardous areas are locations where flammable gases, vapors, or dusts are present, and can ignite or explode if not properly managed.

Chemical engineers must be aware of the potential hazards and take measures to prevent accidents and ensure a safe working environment.

1.0 Industrial Safety and Analysis

Industrial safety involves identifying and mitigating hazards in the workplace. Analysis of potential hazards is crucial to preventing accidents and ensuring a safe working environment. Chemical engineers must be able to identify potential hazards, assess the risks, and implement measures to mitigate them.

1.1 Definition of Hazardous Area

A hazardous area is a location where fire or explosion hazards may exist due to the presence of flammable gases, vapors, dusts, or fibers in the air in sufficient quantities. These areas are common in chemical plants, oil refineries, pharmaceutical industries, and other process industries.

1.2 Examples of Hazardous Areas

- Oil and gas processing units
- Chemical reactors and distillation columns
- Paint shops and storage tanks
- Areas handling combustible dust (e.g., grain elevators)

2. Classification Hazardous Area

2.1 Zone Classification (IEC/ATEX Standard)

Hazardous areas are classified based on the presence of flammable gases, vapors, or dusts. The classification system helps to determine the level of protection required for electrical equipment and other installations.

Zone 0: Locations where flammable gases or vapors are present continuously or for long periods.

A continuous presence of explosive atmosphere (>1000 hrs/year)

Zone 1: Locations where flammable gases or vapors are likely to be present during normal operation.

Occasional presence during normal operations (10–1000 hrs/year)

Zone 2: Locations where flammable gases or vapors are not likely to be present during normal operation, but may be present in abnormal conditions. Rare presence under abnormal conditions (<10 hrs/year)

2.2 Division Classification (NEC/NFPA Standard) (National Electrical Code / National Fire Protection Association)

Division 1: Hazard present during normal operations

Division 2: Hazard present only under abnormal conditions.

3. Sources of Ignition in Hazardous Areas

Electrical equipment (sparks, heat), Static electricity, Open flames or hot surfaces, Frictional heat, Mechanical impact, and Lightning

4. Control Measures in Hazardous Areas (Safety in Hazardous Areas)

- i. Intrinsically Safe Equipment: Devices that are designed to operate at energy levels that are too low to cause ignition.
- ii. Explosion-Proof Equipment: Enclosures that contain any explosion within the device and prevent its spread.
- iii. Purging and Pressurization: Using inert gases or clean air to keep flammable gases out of enclosures.
- iv. Area Ventilation: Ensuring adequate ventilation to dilute and disperse hazardous gases.
- v. Electrical Equipment: Electrical equipment must be designed and installed to prevent ignition of flammable gases or vapors.
- vi. Ventilation: Adequate ventilation is essential to prevent the accumulation of flammable gases or vapors.

5.0 Industrial Safety and Analysis

Industrial Safety: Industrial safety refers to the management of all operations and events within an industry to protect its employees and assets by minimizing hazards, risks, accidents, and near misses.

Safety Measures; Several safety measures can be implemented to prevent accidents in hazardous areas:

- i. Electrical Equipment: Electrical equipment must be designed and installed to prevent ignition of flammable gases or vapors.
- ii. Ventilation: Adequate ventilation is essential to prevent the accumulation of flammable gases or vapors.
- iii. Personal Protective Equipment: Personal protective equipment (PPE) such as respirators, gloves, and safety glasses must be worn by personnel working in hazardous areas.
- iv. Training: Personnel working in hazardous areas must be trained on the potential hazards and safety measures.

5.1 Hazard Identification and Risk Assessment

Hazard identification and risk assessment are critical steps in ensuring industrial safety. Chemical engineers must be able to identify potential hazards, assess the risks, and implement measures to mitigate them.

□ **HAZOP:** Hazard and Operability (HAZOP) studies are a systematic approach to identifying potential hazards and evaluating their risks.

□ **Risk Assessment:** Risk assessment involves evaluating the likelihood and potential consequences of a hazard.

Case Studies

Several case studies can be used to illustrate the importance of safety in hazardous areas, example;

□ **Flixborough Disaster:** The Flixborough disaster in 1974 was a major industrial accident that occurred due to a leak of cyclohexane vapor.

□ Bhopal Disaster: The Bhopal disaster in 1984 was a major industrial accident that occurred due to a leak of methyl isocyanate.

5.2 Safety Analysis Techniques.

5.2.1 Hazard and Operability Study (HAZOP); A systematic review technique to identify process hazards and operability problems.

5.2.2 Fault Tree Analysis (FTA); Top-down approach to find causes of system failures.

5.2.3 Event Tree Analysis (ETA); Forward-looking method to assess possible outcomes of an initiating event.

5.2.4 Failure Mode and Effects Analysis (FMEA); Bottom-up method to evaluate failure modes and their consequences.

5.2.5 Risk Assessment; Identification, evaluation, and estimation of the levels of risk involved in a situation.

5.3 Safety Management Systems (SMS); Framework for managing safety risks with components like: Policy and Objectives, Risk Management, Safety Assurance and Safety Promotion.

6. Legal and Regulatory Framework

OSHA (Occupational Safety and Health Administration)

NFPA (National Fire Protection Association)

ATEX Directives (Europe)

ISO 45001: Occupational health and safety management

7. Emergency Response Planning

□ Fire safety plans

□ Evacuation procedures

□ Spill containment

□ First aid and medical response

□ Alarm and notification systems

□ Zoning and Barrier Systems: Physical separation and barriers to isolate hazardous areas.

□ Use of flame arrestors and gas detectors.

Legal Implications of Various Losses

Loss prevention in chemical engineering involves identifying and mitigating potential losses that can occur in a chemical plant. These losses can have significant legal implications, including:

□ Financial Losses: Financial losses can occur due to damage to equipment, loss of production, and compensation claims.

□ Environmental Losses: Environmental losses can occur due to spills, leaks, or other releases of hazardous materials.

□ Human Losses: Human losses can occur due to accidents or injuries in the

workplace.

Legal Framework

The legal framework for loss prevention in chemical engineering includes:

- Occupational Safety and Health Administration (OSHA)*: OSHA regulations provide guidelines for workplace safety and health.
- Environmental Protection Agency (EPA): EPA regulations provide guidelines for environmental protection.
- National Fire Protection Association (NFPA): NFPA regulations provide guidelines for fire safety.

Safe Work Practices

Safe work practices are essential in preventing losses in chemical plants. These practices include:

- Personal Protective Equipment (PPE): PPE such as hard hats, safety glasses, and gloves must be worn by personnel working in hazardous areas.
- Lockout/Tagout: Lockout/tagout procedures must be followed to ensure that equipment is safely de-energized before maintenance or repair.
- Confined Space Entry: Confined space entry procedures must be followed to ensure safe entry into confined spaces.

Fire Prevention and Fire-Fighting Systems

1.0 Introduction

Fire prevention and fire-fighting systems are critical components of loss prevention in chemical engineering. Fires can have devastating consequences, including loss of life, property damage, and environmental harm. Chemical engineers must be aware of the potential fire hazards and take measures to prevent and mitigate fires.

1.1 Fire Prevention

Fire prevention involves identifying and mitigating potential fire hazards. This includes:

- Fire Hazards: Identifying potential fire hazards, such as flammable liquids, gases, and solids.
- Ignition Sources: Identifying potential ignition sources, such as electrical equipment, hot surfaces, and open flames.
- Fire Protection Measures: Implementing fire protection measures, such as fire-resistant materials, fire suppression systems, and emergency shutdown systems.

1.2 Fire-Fighting Systems

Fire-fighting systems are designed to detect and suppress fires quickly and effectively. This includes:

- Fire Detection Systems: Fire detection systems, such as smoke detectors and heat detectors, detect fires and alert personnel.
- Fire Suppression Systems: Fire suppression systems, such as sprinkler systems and clean agent systems, suppress fires and prevent them from spreading.
- Fire Extinguishers: Fire extinguishers, such as portable extinguishers and

wheeled extinguishers, provide a means of manually suppressing fires.

2.0 Types of Fire Suppression Systems

There are several types of fire suppression systems, including:

- Water-Based Systems: Water-based systems, such as sprinkler systems, use water to cool and smother fires.
- Clean Agent Systems: Clean agent systems use a clean agent, such as a gas or liquid, to suppress fires without leaving a residue.
- Foam Systems: Foam systems use a foam agent to smother fires and prevent them from spreading.

2.1 Definition of Fire: Fire is a rapid oxidation process accompanied by the evolution of heat and light in varying intensities. It requires three elements to occur, this is known as the Fire Triangle:

Fuel: Any combustible material.

Oxygen: Usually from the air.

Heat: Sufficient to ignite the fuel (ignition source).

2.2 Importance of Fire Prevention

- Protects life and property.
- Reduces financial losses due to downtime and damage.
- Ensures legal compliance and safety standards.
- Prevents environmental pollution and liability.

3.0 Fire Hazards in Process Industries:

Common Causes of Industrial Fires

- Leakage of flammable gases or liquids.
- Electrical faults (short circuits, overloads).
- Static electricity and sparks.
- Hot surfaces or open flames.
- Mechanical friction.
- Improper handling or storage of flammable materials.

1.7 Classes of Fire (Based on Fuel Type)

Class A: Combustibles (wood, paper, textiles)

Class B: Flammable liquids/gases (oil, gasoline)

Class C: Electrical fires

Class D: Combustible metals (magnesium, sodium)

Class K/F: Cooking oils and fats (mainly kitchen fires)

4.0 Fire Prevention Strategies:

Engineering Controls;

- Explosion-proof equipment. Proper ventilation in hazardous areas
- Automatic shut-off systems

- Gas detection systems
- Static discharge control

Administrative Controls

- Safety policies and procedures
- Permit to work (PTW) for hot work
- Training and awareness programs
- Regular maintenance and inspections

Good Housekeeping Practices

- Keeping work areas clean and free of combustible materials.
- Proper storage of flammable substances.
- Prompt disposal of waste and spills.

5.0 Fire Detection Systems

Types of Fire Detectors

Smoke Detectors: Ionization or photoelectric type.

Heat Detectors: Fixed temperature and rate-of-rise types.

Flame Detectors: Infrared (IR), ultraviolet (UV), or combined.

Gas Detectors: Detect leaks of flammable gases.

Alarm Systems

Audible and visual alarms.

Centralized monitoring through control panels.

5.1 Fire-Fighting Systems

Portable Fire Extinguishers

Water-based: For Class A fires.

Foam: For Class B fires.

CO2: For Class B and C fires.

Dry chemical powder: Versatile for A, B, C classes.

Wet chemical: For Class K/F fires.

5.2 Fixed Fire Suppression Systems

Sprinkler Systems: Automatic, water-based.

Deluge Systems: Release large quantities of water.

Foam Systems: Used for flammable liquid fires.

Gas Suppression Systems:

Inert gases (e.g., argon)

Chemical agents (e.g., FM-200)

5.3 Hydrant and Hose Systems

External water supply systems with hydrants.

Fire hoses connected to hydrants or internal reels.

5.4 Fire Brigades and Emergency Response Teams

Trained personnel on-site.

Equipped with PPE, breathing apparatus, and firefighting tools.

6. Fire Safety Planning and Regulations

6.1 Fire Risk Assessment

- ☐ Identify fire hazards.
- ☐ Evaluate risks and existing controls.
- ☐ Recommend improvements.

6.2 Fire Safety Plan Components

- ☐ Emergency evacuation routes.
- ☐ Fire alarm and communication systems.
- ☐ Roles and responsibilities.
- ☐ Training and drills.

6.3 Regulatory Framework

- ☐ OSHA Fire Safety Standards (29 CFR 1910)
- ☐ NFPA Codes (e.g., NFPA 10, NFPA 13)
- ☐ ISO 45001
- ☐ Local fire safety regulations

Design Considerations

When designing fire-fighting systems, several factors must be considered, including:

- ☐ Hazard Classification: The hazard classification of the area, including the type of fire hazards present.
- ☐ System Design: The design of the fire-fighting system, including the type of suppression agent and the piping and distribution system.
- ☐ System Maintenance: The maintenance requirements of the fire-fighting system, including regular inspections and testing.